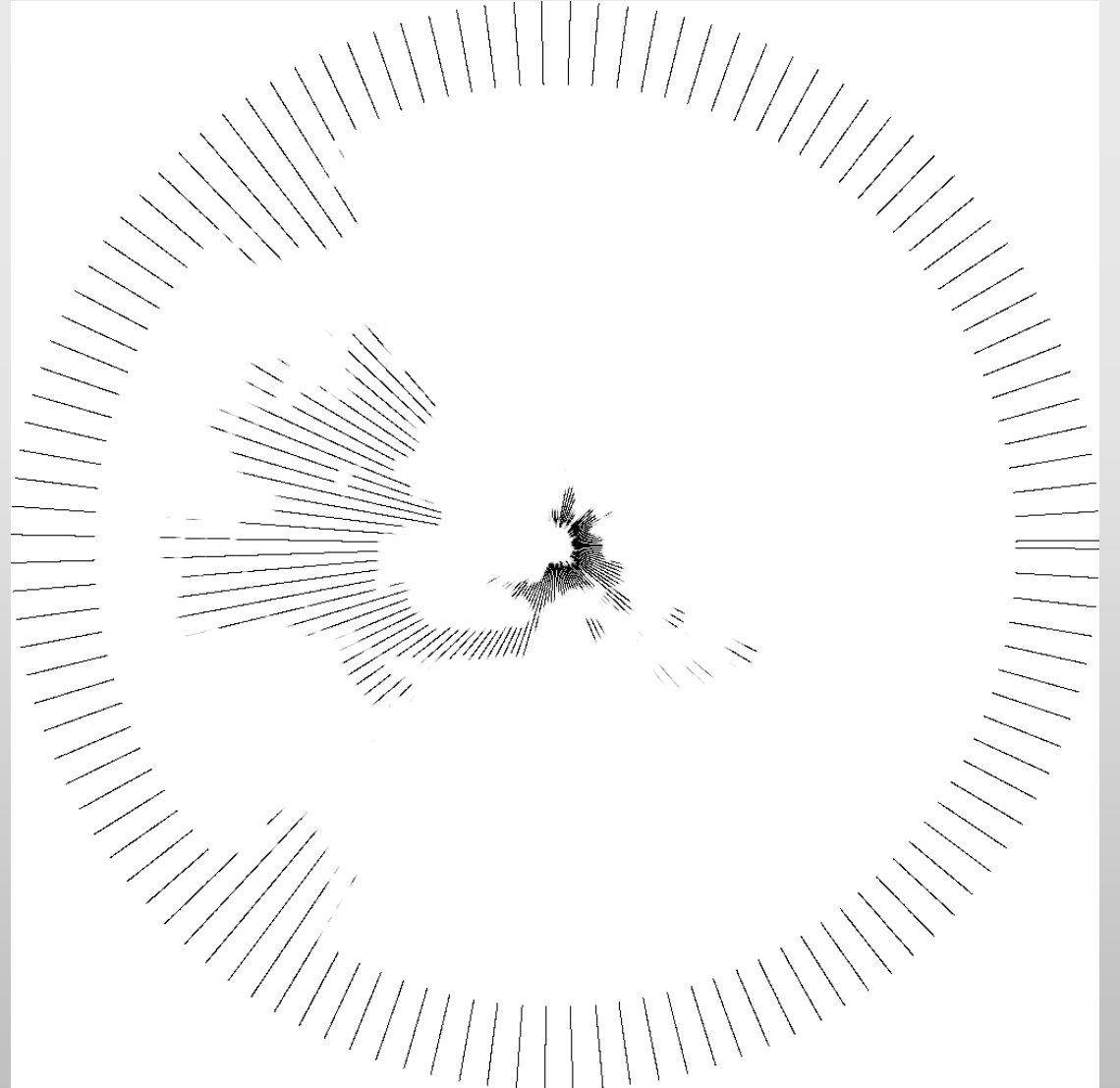


Image Processing and DBSCAN

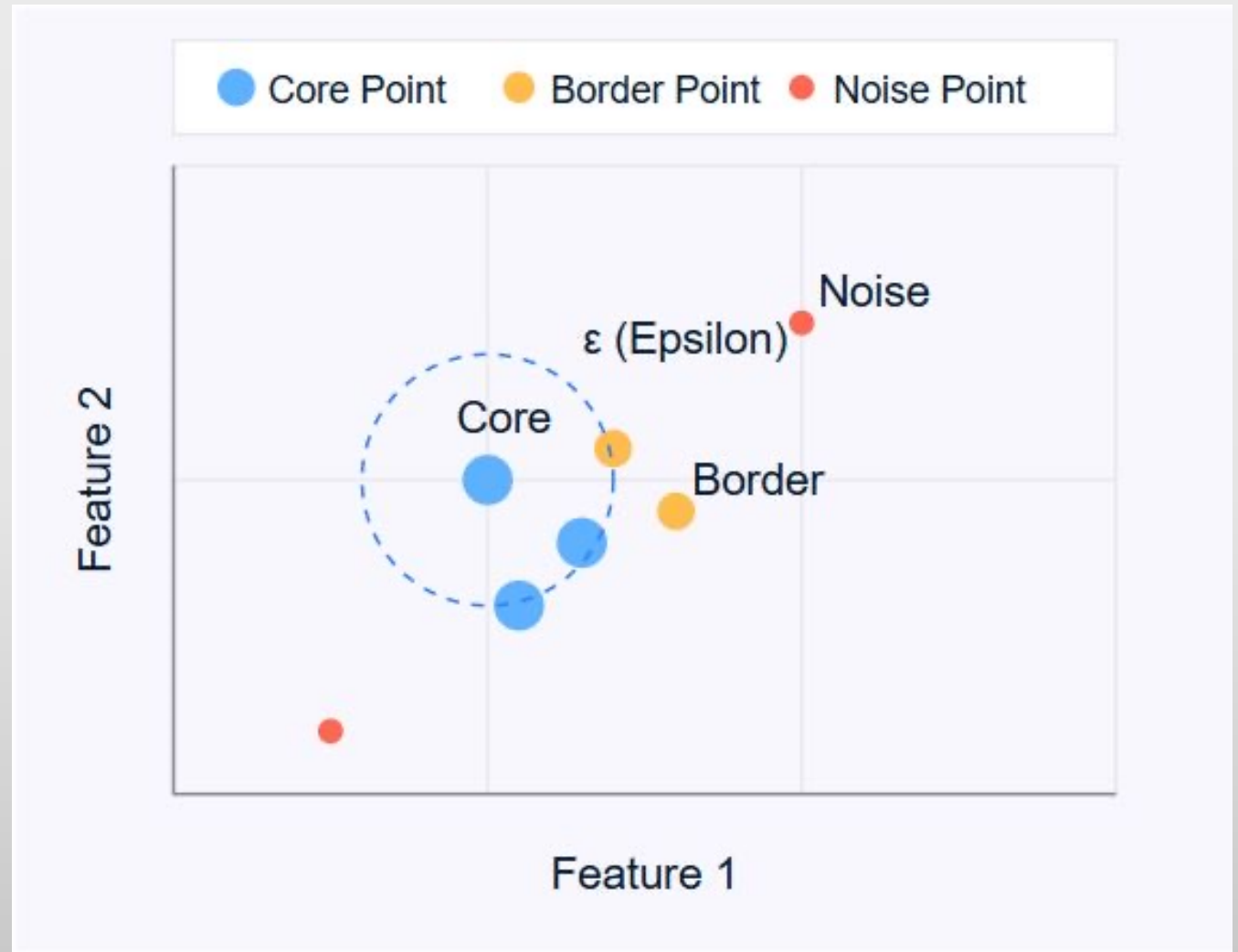
Sam Cancilla 9-2-2025



What is DBSCAN?

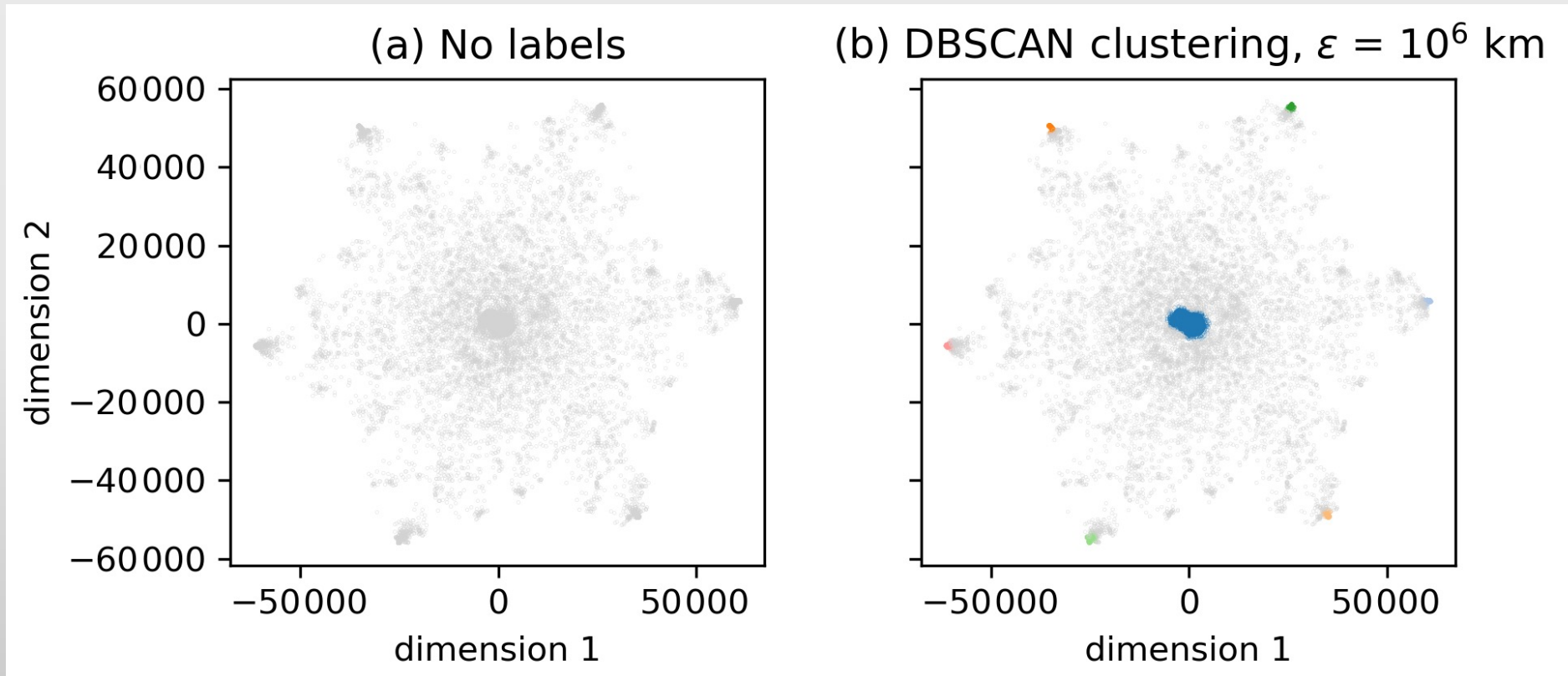
DBSCAN – (not my figure)

- Density Based Spatial Clustering of Applications with Noise (DBSCAN)
- Unsupervised learning method
 - Can feed it a lot of data sets
- No need to disclose cluster count before hand
 - Min data points
 - Epsilon value
- Identifies noise and outlier points



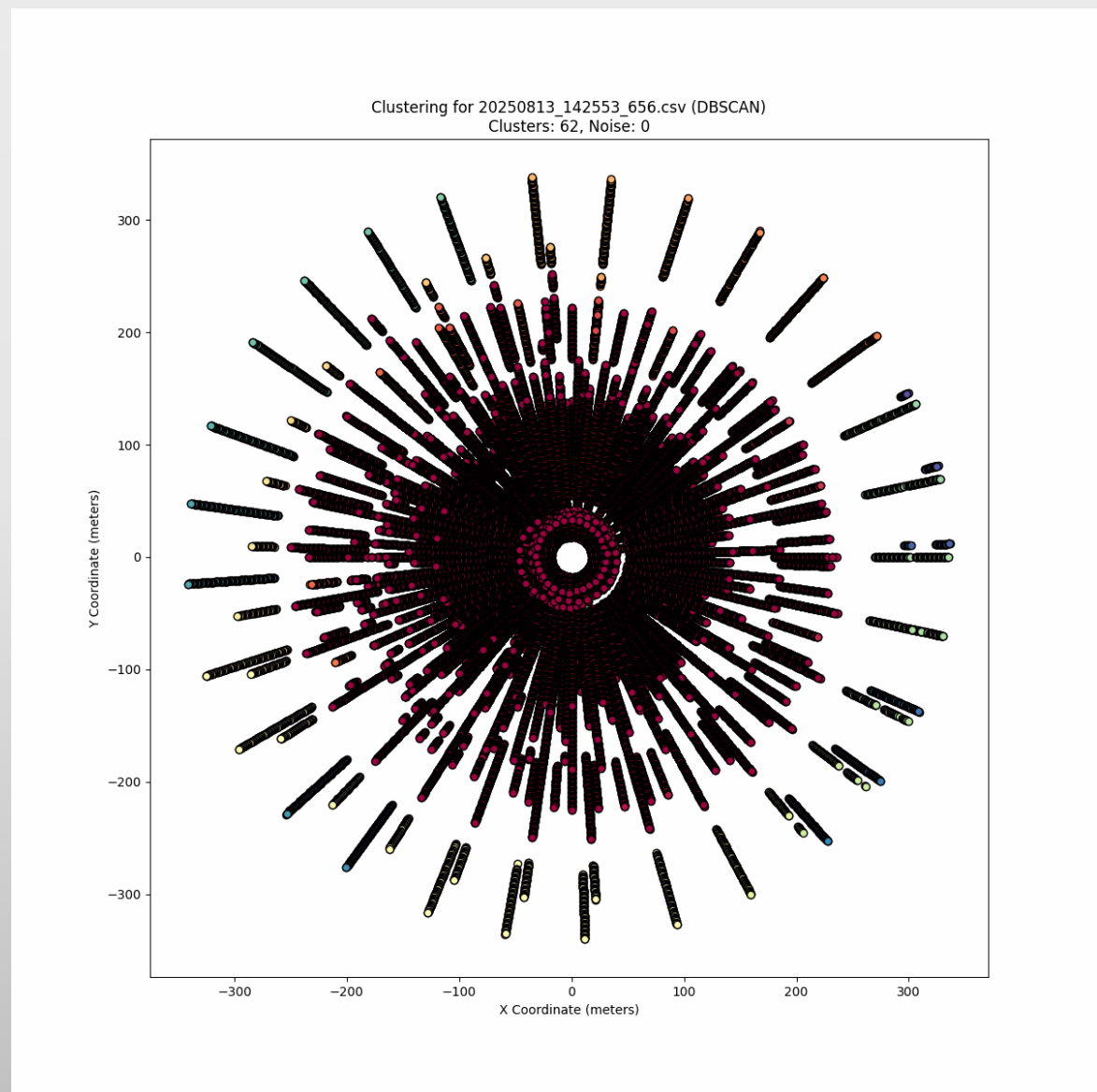
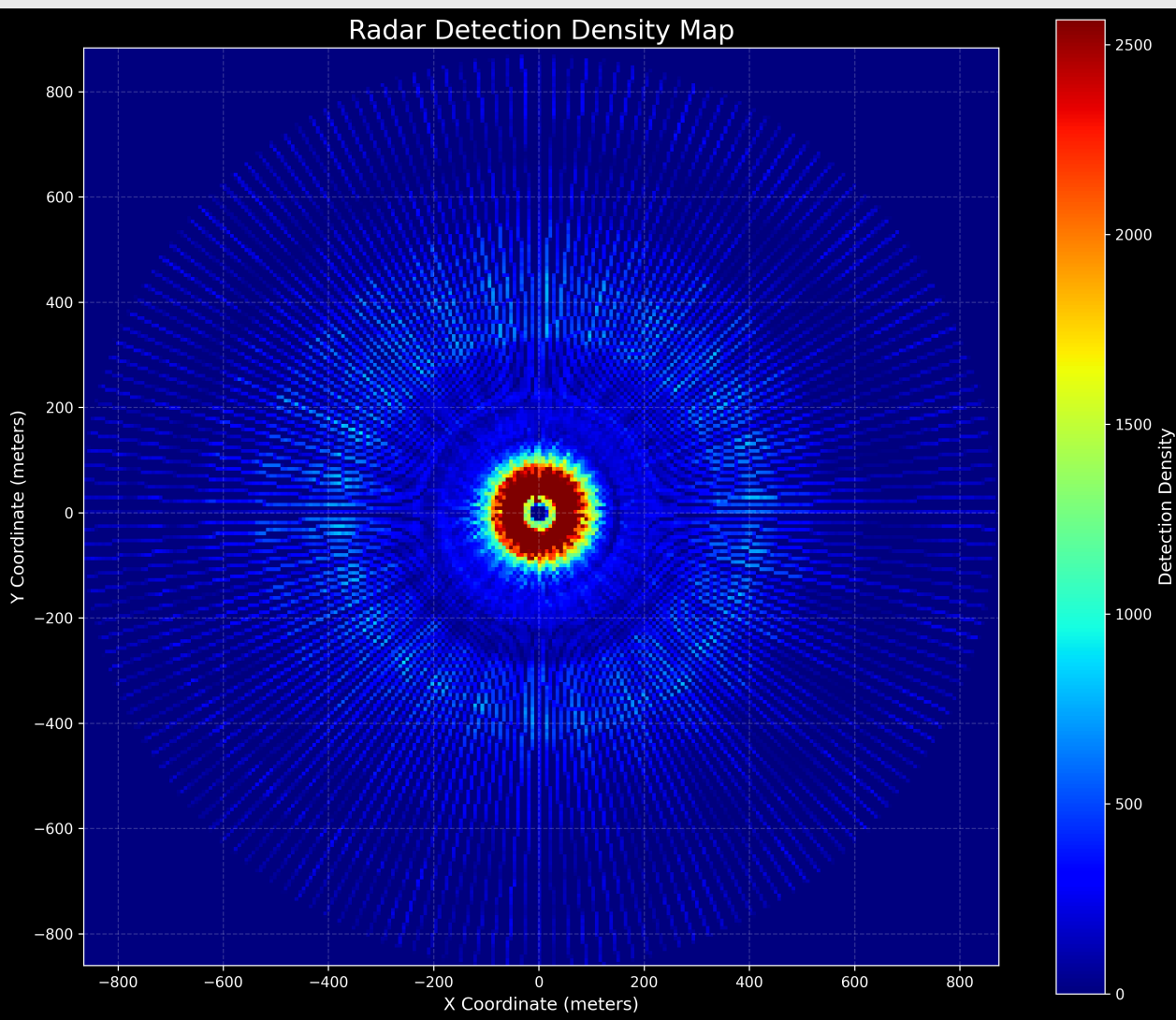
[This Photo](#) by Rajesh Kumar is licensed under [CC BY](#)

DBSCAN in Action – (not my figure)



[This Photo](#) by Unknown Author is licensed under [CC BY](#)

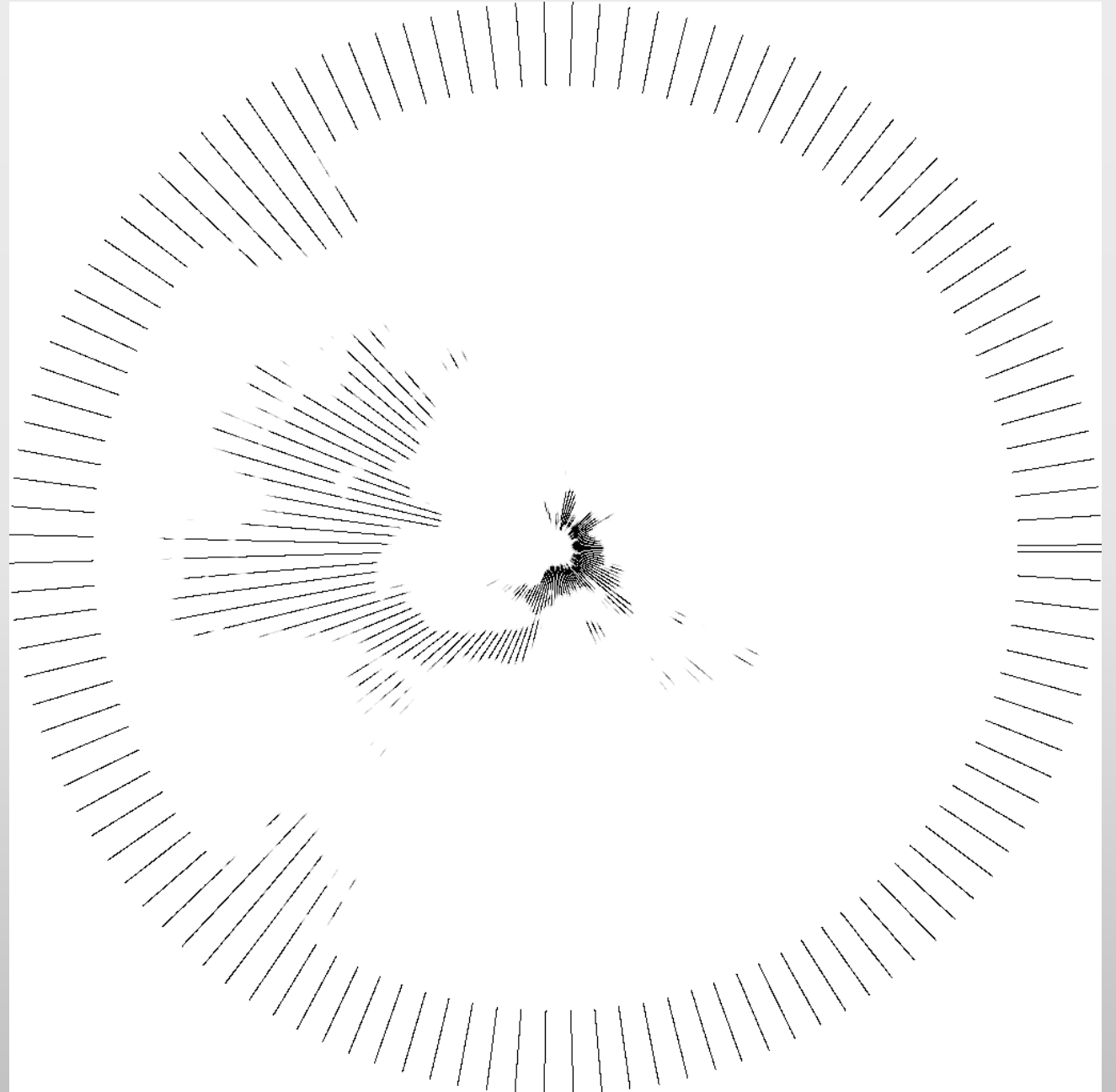
DBSCAN on our Data



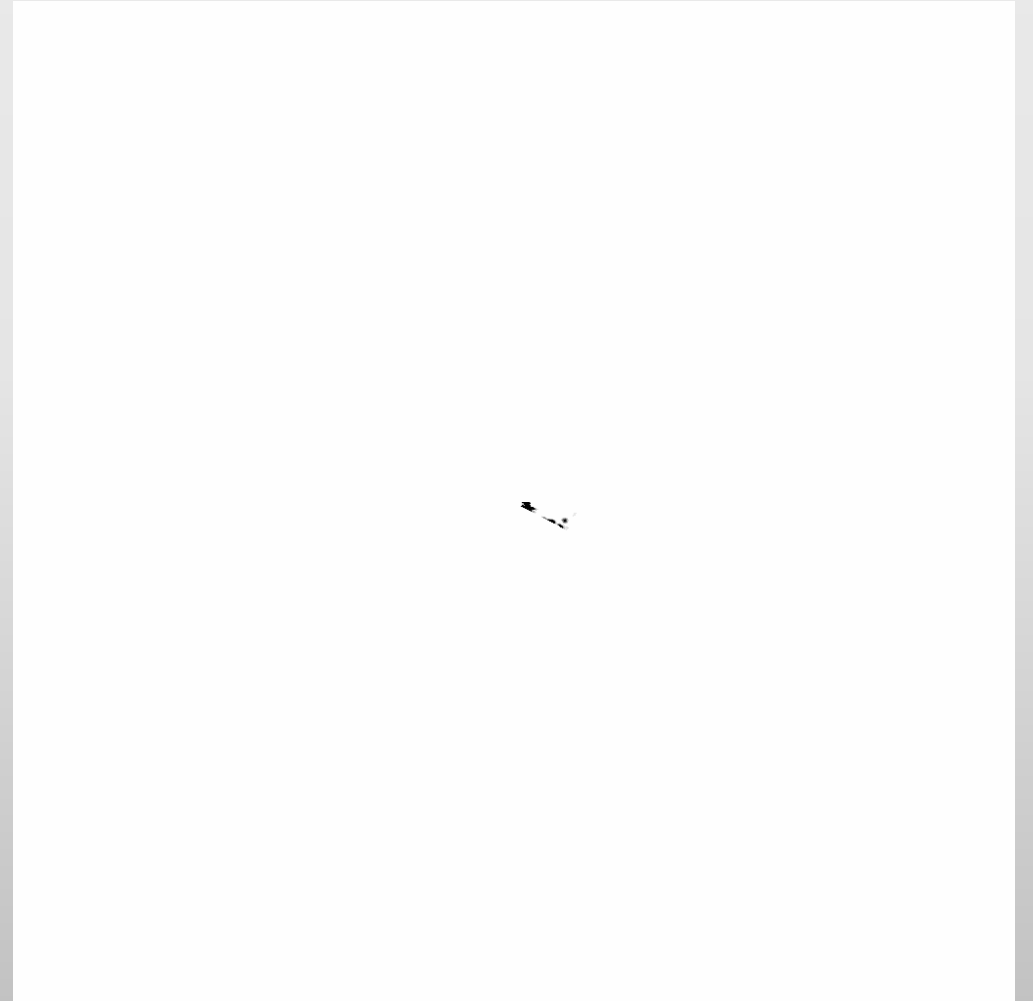
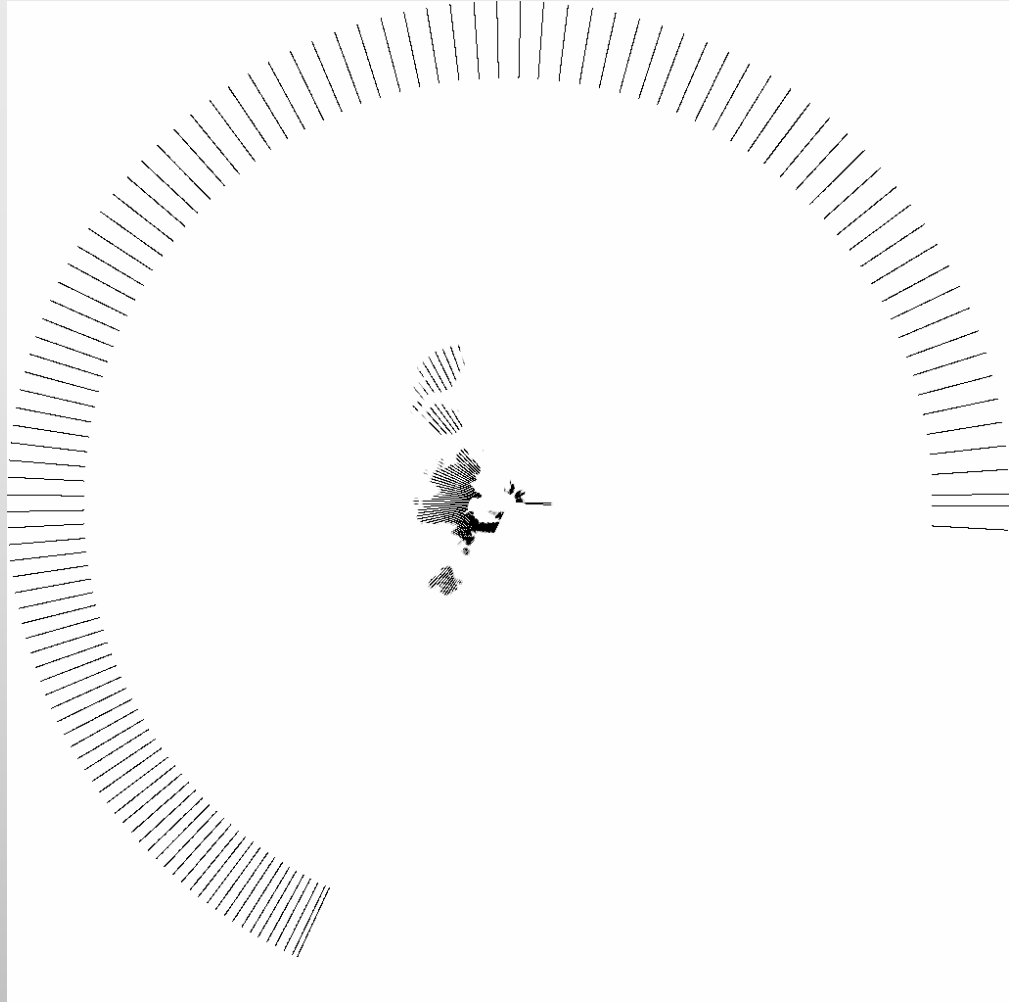
Challenges

Reading the Data

- During the first iteration of DBSCAN the program was reading the CSV data incorrectly
- Revised the program to fix the orientation issues
- Implemented DBSCAN onto the newly orientated data



Side by Side Comparison



What is Happening?

- The data is in a correct orientation as we saw
- It seems like DBSCAN is finding clusters
- Some clusters are being found at varying Azimuths
- You can see the points in the clusters are relatively close to each other
- Range is 0
 - Can implement a domain

```
Clusters found in 20250813_150216_463.csv:
```

```
Cluster 0: 4 points
```

```
(Azimuth: 0.09°, Range: 0.00)
```

```
(Azimuth: 2.99°, Range: 0.00)
```

```
(Azimuth: 5.89°, Range: 0.00)
```

```
(Azimuth: 8.87°, Range: 0.00)
```

```
Cluster 1: 6 points
```

```
(Azimuth: 11.77°, Range: 0.00)
```

```
(Azimuth: 14.76°, Range: 0.00)
```

```
(Azimuth: 17.66°, Range: 0.00)
```

```
(Azimuth: 20.56°, Range: 0.00)
```

```
(Azimuth: 23.46°, Range: 0.00)
```

```
... and 1 more points.
```

```
Finished processing 20250813_150216_463.csv in 0.52 seconds.
```

```
Processing and clustering 20250813_150217_679.csv...
```

Moving forward

- Continue to revise and implement DBSCAN to the data
- Maybe implement a domain
- Implement tracking to fixed DBSCAN data
- Run and compare DBSCAN tracking to ML tracking
- Time based DBSCAN. 3d?



The background is a vibrant blue with a repeating pattern of colorful speech bubbles. Each bubble is a different color (red, yellow, pink, white) and contains a large, bold, dark blue question mark. The bubbles are scattered across the entire frame, creating a sense of many questions being asked.

Thank you.

Any Questions?